WASL · BACKEND ENGINEERING

Wasl Backend — Complete Architecture & Technology Guide

Folder structure, stack, NestJS lifecycle, modules, APIs, data, queues, and operations

> **Priority section:** Repository-wide reference generated from actual source and configuration

Version	1.0
Date	14 October 2026
Repository	wasl-backend
Runtime	Node.js 24+, NestJS 11 + Express, PostgreSQL, Redis, BullMQ
Classification	Internal engineering documentation
Audience	Backend engineers, architects, QA, DevOps, technical leads

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Wasl Backend — Complete Architecture and Technology Guide

Document classification: Internal engineering reference
Repository: `wasl-backend` (JavaScript ESM, Node.js ≥ 24)
Generated: 2026-10-12
Scope: Architecture, folder structure, technology stack, request flows, business modules, API inventory, configuration, database, queues, integrations, testing, operations, risks.

1 Document purpose and scope

This guide describes the **actual** Wasl backend as implemented in this repository. It is intended for engineers onboarding from beginner to advanced level: what each major folder does, how NestJS and domain modules interact, and how HTTP requests, database access, queues, and webhooks flow through the system.

Companion artifacts (full machine-readable inventories):

File	Contents
Wasl_Backend_Source_File_Inventory.csv	Every `.js` / `.mjs` / `.cjs` under `src/`, `tests/`, `test/`, `scripts/` with a **layer** tag
Wasl_Backend_Nest_Endpoint_Inventory.md	**159** HTTP routes from Nest `@Get/@Post/...` decorators
Wasl_Backend_Environment_Variables.md	Environment keys from `src/app/config/schema.js`

Excluded from source inventory: `node_modules/`, build output, secrets in `.env`, and generated lockfile trees. **925** application-related source paths are listed in the CSV.

Verification note: Runtime behavior described here is derived from static analysis of source and configuration unless explicitly marked as execution-verified.

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3 1. Repository overview

3.1 1.1 What Wasl backend is

Wasl is a **multi-tenant commerce and channel management platform**. The backend exposes REST APIs for tenants, products, inventory, channels, marketplaces, orders, fulfillment (shipments, cancellations, returns), webhooks, audit, authorization, and **merchant compatibility** layers (ChannelEngine-style and StockConnect integration paths).

3.2 1.2 Architectural style

The codebase follows a **modular monolith** (see `docs/decisions/ADR-001-modular-monolith.md`):

- **Domain modules** under `src/modules/*` — use cases, repositories, Zod schemas, legacy Fastify-style route modules (still present for wiring/tests in places).
- **HTTP adapter** under `src/nest/*` — NestJS 11 controllers/services on **Express** (`@nestjs/platform-express`).
- **Composition root** — `src/app/bootstrap/create-application.js` wires domain modules and passes a `coreDomain` object into Nest.
- **Infrastructure** — `src/infrastructure/*` (Postgres, Redis, BullMQ, S3, HTTP client, auth primitives).
- **Workers** — separate process `src/workers/main.js` consuming BullMQ jobs (catalog sync, marketplace lifecycle, webhook delivery, etc.).

3.3 1.3 Process model

Process	Entry	Role
API	`src/app/main.js`	Loads config → infrastructure → `createApplication()` → Nest HTTP listen
Worker	`src/workers/main.js`	Same config/infra pattern; registers queue handlers + optional observability HTTP
Migrations CLI	`src/app/bootstrap/migrate-cli.js`	Applies SQL in `src/infrastructure/postgres/migrations/`
Standalone Nest bootstrap	`src/nest/bootstrap.mjs`	Alternate dev entry (`npm run dev:nest`); production path uses `main.js`

3.4 1.4 Repository statistics (scan)

Metric	Count
Directories (excluding `node_modules`, `.git`)	280
Files (same exclusion)	1491

Metric	Count
`src/**/*.js` in inventory	925
SQL migrations	47
Domain modules (`src/modules/`)	22
Nest feature areas (`src/nest/`)	27
Nest decorator routes (static scan)	159
Declared npm dependencies reviewed	42

4 2. Folder tree and layering

4.1 2.1 Top-level tree (depth ≤ 2)

```
wasl-backend/
├── .dependency-cruiser.cjs
├── .dockerignore
├── .env
├── .env.example
├── .gitignore
├── .local/
│   └── bin/
│       └── minio.exe
├── .swcrc
├── docker-compose.yml
├── Dockerfile
├── docs/
│   ├── .openapi-snapshot.json
│   ├── .pdf-pages/
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│   │   ├── page-002.png
│   │   ├── page-003.png
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│   │   ├── ADR-008-multi-tenancy.md
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├── integration-phase4c.log
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├── integration-phase6-pre.log
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│   │   └── health.module.js
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│   │   ├── auth/
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│   │   ├── observability/
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```

```
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│   ├── integrations/
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│   │   ├── returns/
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│   │   ├── logging/
│   │   ├── metrics/
│   │   ├── money/
│   │   ├── pagination/
│   │   ├── persistence/
│   │   ├── queue/
│   │   ├── rate-limit/
│   │   ├── security/
│   │   ├── stockconnect/
│   │   ├── storage/
│   │   └── validation/
│   └── workers/
│       ├── bootstrap/
│       ├── create-integration-event-consumers.js
│       ├── handlers/
│       ├── main.js
│       └── observability/
├── test/
│   └── nest/
│       ├── catalog-modules.test.js
│       ├── channels-marketplaces.test.js
│       ├── core-modules.test.js
│       ├── feature-modules.test.js
│       ├── foundation.test.js
│       ├── helpers/
│       ├── infrastructure.test.js
│       ├── legacy-compatibility.test.js
│       ├── legacy-orders.test.js
│       ├── marketplace-webhooks.test.js
│       ├── order-adjacent-apis.test.js
│       ├── orders.test.js
│       └── setup.js
├── tests/
│   ├── helpers/
│   │   └── test-marketplace-webhook-adapter.js
│   ├── integration/
│   │   ├── amazon-catalog-sync-mappings.test.js
│   │   ├── api-keys.test.js
│   │   ├── auth-flow.test.js
│   │   ├── auth-helpers.js
│   │   ├── auth-rate-limit.test.js
│   │   ├── authorization.test.js
│   │   ├── channel-default-stock-location.test.js
│   │   ├── commerce-foundation.test.js
│   │   ├── commerce-inventory.test.js
│   │   ├── commerce-tenant-isolation.test.js
│   │   ├── compatibility-cancellations-read.test.js
│   │   ├── compatibility-cancellations.test.js
│   │   ├── compatibility-catalog.test.js
│   │   ├── compatibility-channel-fulfilled-orders.test.js
│   │   ├── compatibility-channel-orders.test.js
│   │   ├── compatibility-external-id-enrichment.test.js
```

```
│   │   ├── compatibility-helpers.js
│   │   ├── compatibility-inbound-external-id-resolution.test.js
│   │   ├── compatibility-orders-acknowledge.test.js
│   │   ├── compatibility-orders-new.test.js
│   │   ├── compatibility-orders.test.js
│   │   ├── compatibility-rate-limit.test.js
│   │   ├── compatibility-return-mutations.test.js
│   │   ├── compatibility-returns-read.test.js
│   │   ├── compatibility-returns.test.js
│   │   ├── compatibility-shipment-tracking.test.js
│   │   ├── compatibility-shipments-read.test.js
│   │   ├── compatibility-shipments.test.js
│   │   ├── external-consumer-compatibility-boundary.test.js
│   │   ├── external-id-backfill.test.js
│   │   ├── external-id-mapping-creation-wiring.test.js
│   │   ├── external-integer-id-mapping.test.js
│   │   ├── helpers.js
│   │   ├── http.test.js
│   │   ├── idempotency.test.js
│   │   ├── marketplace-connection-entity-mapping.test.js
│   │   ├── marketplace-lifecycle-worker.test.js
│   │   ├── marketplace-order-ingestion.test.js
│   │   ├── marketplace-order-lifecycle.test.js
│   │   ├── marketplace-webhook-ingestion.test.js
│   │   ├── mfa.test.js
│   │   ├── namshi-catalog-sync-mappings.test.js
│   │   ├── namshi-order-lifecycle.test.js
│   │   ├── nexora-external-integration-readiness.test.js
│   │   ├── node-swc-register.mjs
│   │   ├── noon-catalog-sync-mappings.test.js
│   │   ├── order-command-service.test.js
│   │   ├── outbox-inbox.test.js
│   │   ├── phase4-concurrency.test.js
│   │   ├── phase4-idempotency-mutations.test.js
│   │   ├── phase4-orders.test.js
│   │   ├── phase7-channel-order-ingestion.test.js
│   │   ├── postgres-rls.test.js
│   │   ├── postgres.test.js
│   │   ├── redis.test.js
│   │   ├── retention-cleanup.test.js
│   │   ├── setup.js
│   │   ├── shipment-fulfillment-inventory.test.js
│   │   ├── shopify-catalog-sync-mappings.test.js
│   │   ├── shopify-order-ingestion.test.js
│   │   ├── shopify-order-lifecycle.test.js
│   │   ├── stock-connect-integration.test.js
│   │   ├── stockconnect-ce-catalog-ops.test.js
│   │   ├── stockconnect-ce-catalog.test.js
│   │   ├── stockconnect-ce-e2e-contract.test.js
│   │   ├── stockconnect-ce-foundation.test.js
│   │   ├── stockconnect-ce-invoice.test.js
│   │   ├── stockconnect-ce-production-readiness.test.js
│   │   ├── stockconnect-ce-route-inventory.test.js
│   │   ├── storage.test.js
│   │   ├── tenant-isolation.test.js
│   │   ├── tenant-lifecycle-auth.test.js
│   │   ├── webhook-catalog-expansion.test.js
│   │   ├── webhook-commerce-catalog-expansion.test.js
│   │   ├── webhook-delivery-worker.test.js
│   │   ├── webhook-dispatch-enqueue.test.js
│   │   ├── webhooks-admin-api.test.js
```

```
│   │   ├── webhooks-persistence.test.js
│   │   └── webhooks-tenant-isolation.test.js
│   ├── README.md
│   └── unit/
│       ├── api-keys/
│       ├── architecture/
│       ├── auth/
│       ├── authorization/
│       ├── cancellations/
│       ├── channel-catalog-sync/
│       ├── channels/
│       ├── compatibility/
│       ├── config.test.js
│       ├── errors.test.js
│       ├── events/
│       ├── external-id-mapping/
│       ├── idempotency-fingerprint.test.js
│       ├── identity/
│       ├── logging-redaction.test.js
│       ├── marketplace-order-ingestion/
│       ├── marketplace-webhook-ingestion/
│       ├── marketplaces/
│       ├── money.test.js
│       ├── orders/
│       ├── pagination.test.js
│       ├── postgres-errors.test.js
│       ├── products/
│       ├── request-context.test.js
│       ├── retention-cleanup.test.js
│       ├── retention-config.test.js
│       ├── returns/
│       ├── security/
│       ├── shared/
│       ├── shipments/
│       ├── stockconnect-ce-auth-readers.test.js
│       ├── tenants/
│       ├── validation.test.js
│       ├── webhooks/
│       ├── worker-observability-config.test.js
│       └── workers/
├── vitest.config.js
└── workers/
    └── README.md
```

4.2 2.2 `src/app/` — application bootstrap and cross-cutting

Path	Purpose
`src/app/main.js`	API process entry: config, infra, `createApplication`, outbox publisher, pool metrics, graceful shutdown.
`src/app/bootstrap/create-application.js`	**Composition root**: instantiates all domain modules, marketplace adapters, compatibility, webhooks; returns `httpServer` via `createNestHttpServer`.
`src/app/bootstrap/create-infrastructure.js`	Database pool, Redis, BullMQ queue, S3 storage, logger, metrics, tracing, rate limiter, outbox.

Path	Purpose
`src/app/bootstrap/shutdown.js`	Ordered teardown: readiness, HTTP, outbox, queue, Redis, DB, tracing.
`src/app/config/schema.js`	Zod schema — **single source of truth** for environment variables.
`src/app/config/config.js`	Maps validated env to nested `config` object consumed by modules.
`src/app/http/create-nest-http-server.js`	Creates Nest app; exposes `inject()` for Vitest integration tests (Supertest).
`src/app/observability/readiness.js`	Readiness probes (DB, Redis, queue, storage).
`src/app/register-swc.mjs`	SWC register hook for Nest decorators when running `main.js`.

Why it exists: Keeps framework (Nest) at the edge while preserving a explicit, testable domain wiring layer independent of HTTP.

4.3 2.3 `src/modules/` — domain bounded contexts

Each module typically contains:

- `application/` — use cases and domain services (plain classes/functions).
- `infrastructure/` — Postgres repositories, adapters.
- `presentation/` — Zod schemas and legacy `*.routes.js` (Fastify-era route registration; **HTTP is now primarily Nest**).
- `index.js` — `createXModule(deps)` factory exporting services for composition root.

Used by: `create-application.js`, workers, and Nest services via `CORE_DOMAIN` injection token.

4.4 2.4 `src/nest/` — HTTP adapter (NestJS + Express)

Area	Responsibility
`bootstrap/create-nest-application.js`	`NestFactory.create` with `ExpressAdapter`, Helmet/body limits via `configure-express.js`.
`app.module.js`	Dynamic `buildAppModule(infra, coreDomain)` — registers feature modules and global `AuthGuard` + `GlobalExceptionFilter`.
`domain/core-domain.module.js`	Provides `CORE_DOMAIN` token with wired use cases.
`common/guards/auth.guard.js`	JWT, API key (`X-Api-Key` , `ApiKey`), merchant compat query keys, StockConnect Bearer `nxk_` keys.
`common/filters/global-exception.filter.js`	Maps domain errors to HTTP responses.
`*/*.controller.js`	Route handlers — see endpoint inventory.

Area	Responsibility
`stock-connect-integration/`	`/api/v2/stock-connect/*` integration API (when enabled).
`legacy-compatibility/`, `legacy-orders/`	ChannelEngine / v1 compatibility surfaces.

4.5 2.5 `src/infrastructure/`

PostgreSQL (`pg`), migrations, outbox/inbox, Redis (`ioredis`), BullMQ queue worker runtime, S3 client, fetch HTTP client, JWT (jose), argon2, AES encryptor, OpenTelemetry hooks.

4.6 2.6 `src/workers/`

Queue consumer process: `handlers/queue-job-handlers.js`, marketplace lifecycle wiring, catalog sync, webhook delivery, integration event consumers.

4.7 2.7 `src/shared/` and `src/integrations/`

Cross-cutting utilities (errors, context, auth route prefixes). `src/integrations/stock-connect/` holds StockConnect DTO mappers/schemas separate from Nest controllers.

4.8 2.8 `tests/`, `test/`, `scripts/`

Path	Role
`tests/unit/`	Vitest unit tests (no external services).
`tests/integration/`	Vitest + real Postgres/Redis/MinIO; uses Nest `inject`.
`test/nest/`	Jest tests focused on Nest controllers/guards.
`scripts/check-syntax.mjs`	`npm run build` — parses all `src/**/*.js` (skips `src/nest` per config).
`scripts/audit-route-inventory.mjs`	**Legacy** Fastify route scan — can disagree with Nest inventory.

4.9 2.9 Repository root files (non-`src/`)

Path	Purpose
`package.json`	ESM package manifest, Node ≥ 24, npm scripts for API/worker/tests/migrations.
`package-lock.json`	Locked dependency versions (see §3).
`vitest.config.js`	Unit vs integration Vitest projects; SWC for Nest in integration tests.
`jest.config.js`	Nest Jest suite under `test/nest/`.
`eslint.config.js`	ESLint flat config; **ignores** `src/nest/**` for SWC decorator syntax.

Path	Purpose
`.dependency-cruiser.cjs`	Module boundary rules (`npm run arch:check`).
`Dockerfile`	Multi-stage Node 24 Alpine image; default CMD `node src/app/main.js` ; worker override documented in comments.
`docker-compose.yml`	Local Postgres, Redis (BullMQ-safe), MinIO; optional observability profile.
`.env.example`	Safe template for required env vars (no production secrets).
`docs/`	ADRs, migration notes, Postman/Swagger reports, this architecture guide.
`infrastructure/observability/`	OTel collector and Prometheus configs for compose profile.

4.10 2.10 Per-file inventory policy

Documenting **925** files inline would reduce readability. The CSV inventory lists **every** source path with layer:

- `nest-http` , `domain-module` , `application-bootstrap` , `infrastructure` , `workers` , `shared` , `integrations` , `vitest` , `jest-nest` , `scripts` .

For any path `src/modules/<module>/application/<use-case>.js` , expect a single use case class/function invoked from the module factory or Nest service.

Inventory rows by layer (`1171` total paths in CSV):

Layer	File count
domain-module	693
vitest	220
nest-http	99
shared	70
infrastructure	27
application-bootstrap	17
workers	15
jest-nest	15
scripts	11
integrations	4

Typical file roles within a domain module:

Subfolder	Example	Responsibility
`application/`	`list-orders.js`	Orchestrates authorization + repository calls for one operation.
`infrastructure/`	`postgres-order-repository.js`	SQL queries; maps rows to domain types.
`presentation/`	`order.schemas.js`	Zod request/response contracts shared by Nest and legacy routes.
`presentation/`	`order.routes.js`	Legacy route registration (Fastify-era); not the primary HTTP surface.
`domain/`	entity/value objects	Invariants and pure logic where extracted.

5 3. Technology stack

5.1 3.1 Master dependency table

Technology/package	Verified version	Category	Why we use it	Actual usage locations
`@aws-sdk/client-s3`	3.1136.0	runtime	See §3 Technology stack	`src/`, `tests/`, workers
`@eslint/js`	9.39.5	dev	See §3 Technology stack	`src/`, `tests/`, workers
`@jest/globals`	29.7.0	dev	See §3 Technology stack	`src/`, `tests/`, workers
`@nestjs/common`	11.2.6	runtime	See §3 Technology stack	`src/`, `tests/`, workers
`@nestjs/core`	11.2.6	runtime	See §3 Technology stack	`src/`, `tests/`, workers
`@nestjs/platform-express`	11.2.7	runtime	See §3 Technology stack	`src/`, `tests/`, workers
`@nestjs/testing`	11.2.7	dev	See §3 Technology stack	`src/`, `tests/`, workers
`@opentelemetry/api`	1.9.1	runtime	See §3 Technology stack	`src/`, `tests/`, workers
`@opentelemetry/exporter-trace-otlp-http`	0.57.2	runtime	See §3 Technology stack	`src/`, `tests/`, workers
`@opentelemetry/instrumentation-http`	0.57.2	runtime	See §3 Technology stack	`src/`, `tests/`, workers
`@opentelemetry/instrumentation-ioredis`	0.47.1	runtime	See §3 Technology stack	`src/`, `tests/`, workers
`@opentelemetry/instrumentation-pg`	0.51.1	runtime	See §3 Technology stack	`src/`, `tests/`, workers
`@opentelemetry/resources`	1.30.1	runtime	See §3 Technology stack	`src/`, `tests/`, workers
`@opentelemetry/sdk-node`	0.57.2	runtime	See §3 Technology stack	`src/`, `tests/`, workers
`@opentelemetry/semantic-conventions`	1.43.0	runtime	See §3 Technology stack	`src/`, `tests/`, workers
`@swc-node/register`	1.12.1	runtime	See §3 Technology stack	`src/`, `tests/`, workers

Technology/package	Verified version	Category	Why we use it	Actual usage locations
`@swc/core`	1.16.12	runtime	See §3 Technology stack	`src/`, `tests/`, workers
`@swc/jest`	0.2.39	dev	See §3 Technology stack	`src/`, `tests/`, workers
`argon2`	0.45.1	runtime	See §3 Technology stack	`src/`, `tests/`, workers
`bullmq`	5.81.5	runtime	See §3 Technology stack	`src/`, `tests/`, workers
`dependency-cruiser`	16.10.4	dev	See §3 Technology stack	`src/`, `tests/`, workers
`dotenv`	18.0.1	dev	See §3 Technology stack	`src/`, `tests/`, workers
`eslint`	9.39.5	dev	See §3 Technology stack	`src/`, `tests/`, workers
`eslint-config-prettier`	9.1.2	dev	See §3 Technology stack	`src/`, `tests/`, workers
`express`	4.22.3	runtime	See §3 Technology stack	`src/`, `tests/`, workers
`globals`	15.15.0	dev	See §3 Technology stack	`src/`, `tests/`, workers
`helmet`	8.3.0	runtime	See §3 Technology stack	`src/`, `tests/`, workers
`ioredis`	5.11.1	runtime	See §3 Technology stack	`src/`, `tests/`, workers
`jest`	29.7.0	dev	See §3 Technology stack	`src/`, `tests/`, workers
`jose`	6.2.12	runtime	See §3 Technology stack	`src/`, `tests/`, workers
`otplib`	13.5.0	runtime	See §3 Technology stack	`src/`, `tests/`, workers
`pg`	8.23.0	runtime	See §3 Technology stack	`src/`, `tests/`, workers
`pino`	9.14.0	runtime	See §3 Technology stack	`src/`, `tests/`, workers
`pino-pretty`	13.1.3	dev	See §3 Technology stack	`src/`, `tests/`, workers

Technology/package	Verified version	Category	Why we use it	Actual usage locations
`prettier`	3.9.8	dev	See §3 Technology stack	`src/`, `tests/`, workers
`prom-client`	15.1.3	runtime	See §3 Technology stack	`src/`, `tests/`, workers
`reflect-metadata`	0.2.2	runtime	See §3 Technology stack	`src/`, `tests/`, workers
`rxjs`	7.8.2	runtime	See §3 Technology stack	`src/`, `tests/`, workers
`supertest`	7.3.0	dev	See §3 Technology stack	`src/`, `tests/`, workers
`unplugin-swc`	2.0.0	dev	See §3 Technology stack	`src/`, `tests/`, workers
`vitest`	3.2.7	dev	See §3 Technology stack	`src/`, `tests/`, workers
`zod`	3.25.76	runtime	See §3 Technology stack	`src/`, `tests/`, workers

5.2 3.2 Core runtime and framework

- **Node.js** ≥ 24 (`package.json#engines`) — native `--env-file`, ESM (`"type": "module"`).
- **NestJS 11** — modules, guards, filters, DI for HTTP layer only.
- **Express 4** — underlying HTTP server via `ExpressAdapter`; custom body parsing in `configure-express.js`.
- **SWC** — decorator compilation for Nest in runtime (`register-swc.mjs`) and Vitest (`unplugin-swc`).

5.3 3.3 Data, queues, storage

- **PostgreSQL 17** (Docker image) — primary store; RLS migrations (`0030_rls_hardening.sql`).
- **Redis 7** — caching, locks, rate limits; **BullMQ** requires `noeviction` (see `docker-compose.yml`).
- **MinIO (S3-compatible)** — object storage via `@aws-sdk/client-s3`.
- **BullMQ** — `src/infrastructure/queue/bullmq-job-queue.js`, `bullmq-worker-runtime.js`.

5.4 3.4 Security and auth

- **jose** — JWT access tokens (`src/infrastructure/auth/jose-access-token-service.js`).
- **argon2** — password hashing.
- **otplib** — MFA TOTP.
- **helmet** — HTTP security headers on Express.
- **Zod** — config and request schema validation.

5.5 3.5 Observability

- **pino** — structured logging.
- **prom-client** — metrics endpoint (when enabled).
- **OpenTelemetry** — optional tracing (`TRACING_ENABLED` , OTLP exporter).

5.6 3.6 Testing and quality

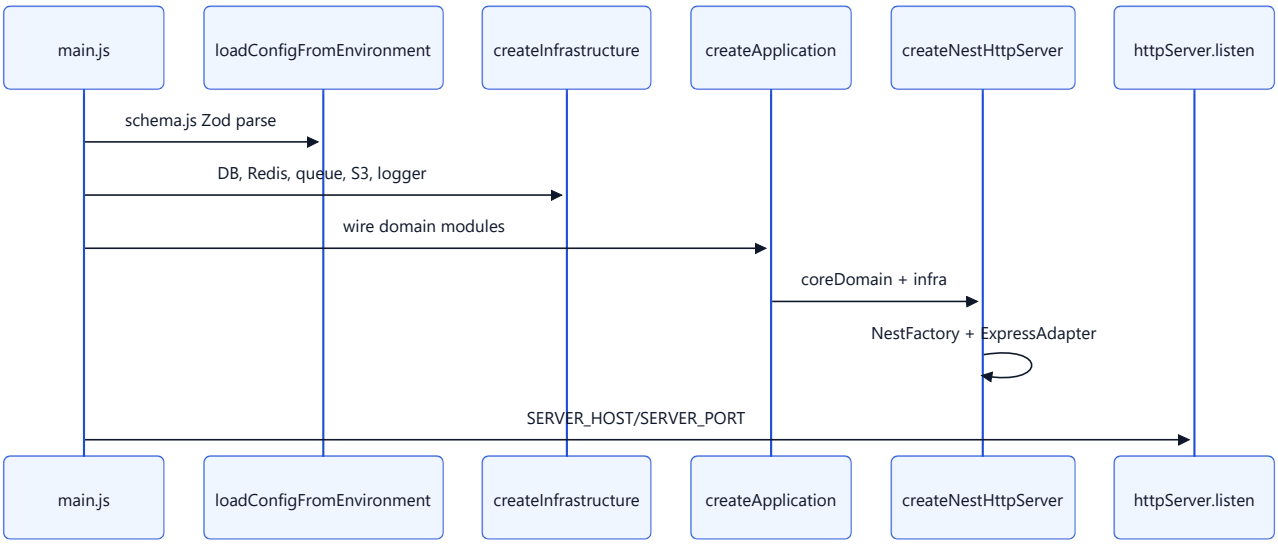
- **Vitest 3** — primary test runner (unit + integration projects).
- **Jest 29** — Nest-focused tests under `test/nest/` .
- **Supertest** — HTTP assertions via `inject` wrapper.
- **ESLint 9 + Prettier + dependency-cruiser** — `npm run verify` .

5.7 3.7 Apparently unused or dev-only

- **dotenv** — devDependency; runtime prefers `--env-file=.env` .
- Packages listed but with **no** `src/` import should be treated as transitive or tooling-only until grep confirms (see risks §13).

6 4. NestJS and Express architecture

6.1 4.1 Bootstrap sequence



Key files: `src/app/main.js`, `src/app/bootstrap/create-application.js`, `src/nest/bootstrap/create-nest-application.js`, `src/nest/app.module.js`.

6.2 4.2 Module registration

When `coreDomain !== null`, `buildAppModule` registers (among others): Auth, Tenants, Authorization, Audit, ApiKeys, MFA, Products, Pricing, Offers, Inventory, Channels, Marketplaces, Webhooks, WebhookSubscriptions, Orders, Cancellations, Shipments, Returns, LegacyOrders, LegacyCompatibility, StockConnectIntegration.

Global providers:

- `APP_FILTER` → `GlobalExceptionFilter`
- `APP_GUARD` → `AuthGuard` (skipped when `coreDomain` is null — e.g. minimal health-only boot)

6.3 4.3 Framework vs domain

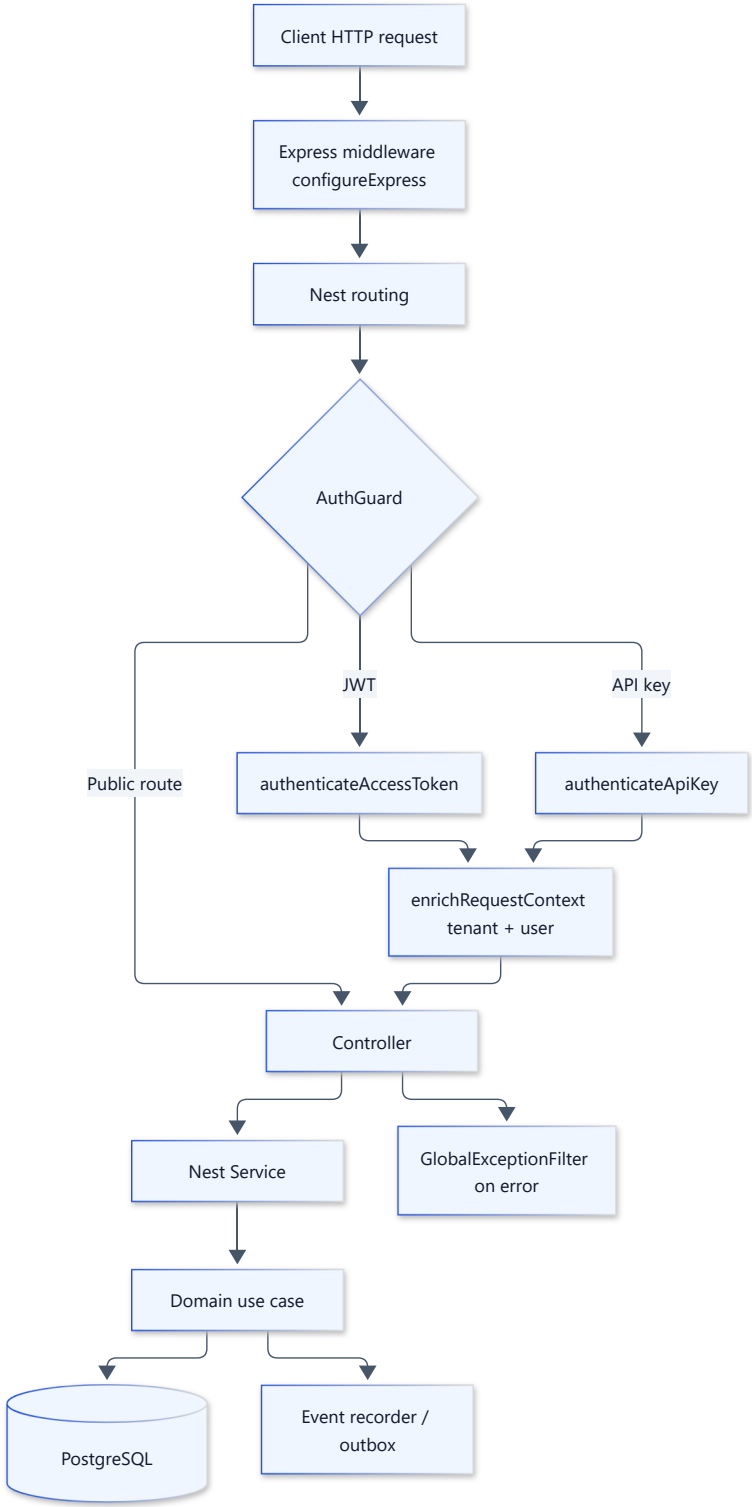
Layer	Location	Responsibility
HTTP routing / status codes	`src/nest/**/*.controller.js`	Map HTTP to service calls
Authorization enforcement	`AuthGuard` + domain `AuthorizationService`	Authentication at edge; permission checks in use cases
Business rules	`src/modules/*/application/*`	Pure domain logic
Persistence	`src/modules/*/infrastructure/*`	SQL via `pg`

6.4 4.4 Fastify migration status

Verified: Production API uses **Nest + Express** (`create-nest-http-server.js`). There is **no** active Fastify server factory in the runtime path described above. Legacy `*.routes.js` files under `src/modules/*/presentation/` remain for historical wiring and tooling; `scripts/audit-route-inventory.mjs` still scans them — treat `docs/API_ROUTE_INVENTORY.md` as **stale** relative to Nest.

7 5. Request and job lifecycles

7.1 5.1 HTTP request (authenticated API)

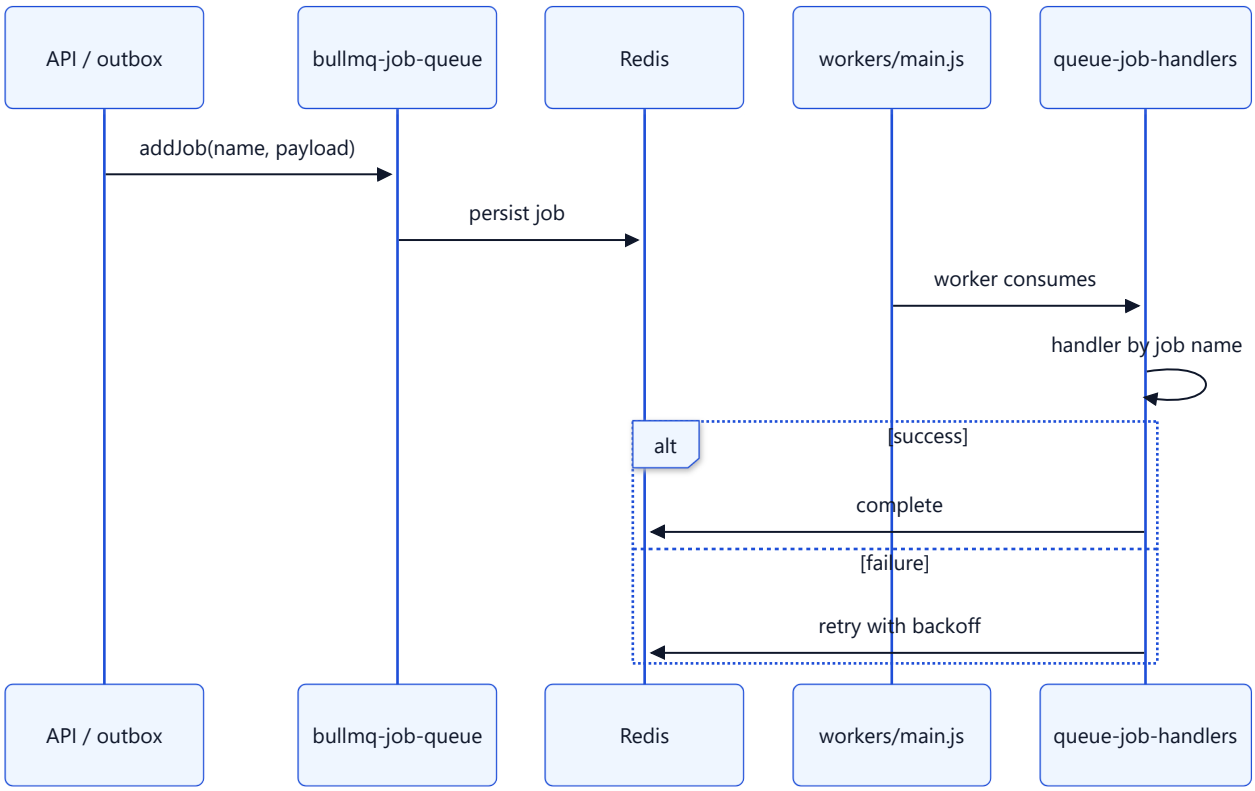


Files: `src/nest/common/guards/auth.guard.js`, `src/modules/identity/application/authenticate-access-token.js`, `src/shared/context/request-context.js`, controllers/services under `src/nest/<feature>/`.

7.2 5.2 Database read/write

Domain repositories use a shared `database` / `queryable` abstraction from infrastructure. Tenant context is set for RLS (see migrations `0001_app_schema_and_tenant_context.sql`, `0030_rls_hardening.sql`). Transactions are coordinated at use-case level where needed.

7.3 5.3 BullMQ job flow



Configuration: `QUEUE_*`, `QUEUE_REDIS_URL` (optional override), `QUEUE_DEFAULT_ATTEMPTS`, `QUEUE_BACKOFF_BASE_MS` in `schema.js`.

7.4 5.4 Marketplace webhook ingress (high level)

Inbound routes live under Nest `webhooks` / marketplace modules and domain `marketplace-webhook-ingestion`. Typical flow: verify signature → normalize payload (Zod) → idempotency/inbox → enqueue processing. See `docs/marketplaces/webhook-ingestion.md` for operational detail aligned with code.

7.5 5.5 Outbound webhooks

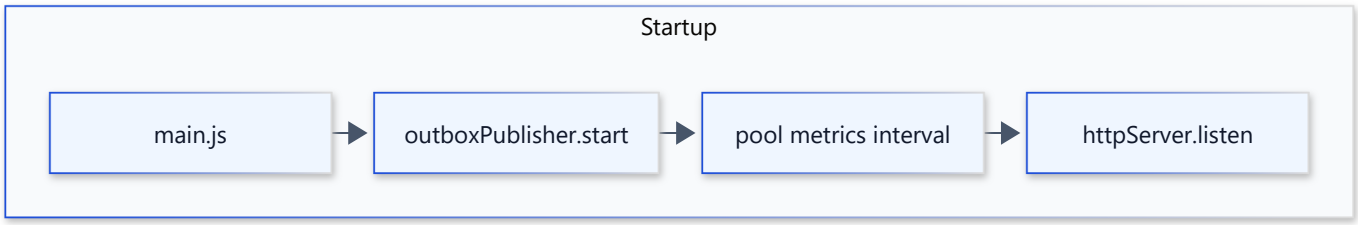
Worker `webhookDeliveryService` loads subscriptions, signs payloads, delivers via `infra.httpClient`, records deliveries in Postgres, respects retry/ `Retry-After` settings (`WEBHOOK_DELIVERY_*` env vars).

7.6 5.6 Health and readiness

`src/nest/health/health.controller.js` exposes health/readiness endpoints (see Nest inventory). `ReadinessService` aggregates probe results from infrastructure components.

7.7 5.7 Application startup (outbox and metrics)

After `createApplication` resolves, `main.js` starts `infra.outboxPublisher` and periodic database pool metrics before listening. On SIGTERM/SIGINT, `gracefulShutdown` drains HTTP, stops outbox/queue consumers, closes Redis and DB pools — see `src/app/bootstrap/shutdown.js`.



8 6. Business modules

Domain modules discovered under `src/modules/`:

8.1 api-keys

Attribute	Detail
Business purpose	Domain logic for **api-keys** (see `src/modules/api-keys/application/` use cases).
Folder	`src/modules/api-keys/` with typical layers: `application/`, `domain/`, `infrastructure/`, `presentation/`.
HTTP (Nest)	`src/nest/api-keys/` Nest module + controller(s).
Factory	`createApiKeysModule` or `index.js` export in `src/modules/api-keys/`.
Persistence	PostgreSQL repositories under `src/modules/api-keys/infrastructure/` where present.
Tests	`tests/unit/**/api-keys*`, `tests/integration/**/api-keys*` (search by module name).

How it fits: Domain modules are composed in `src/app/bootstrap/create-application.js` and injected into Nest via `CoreDomainModule` (`src/nest/domain/`). Controllers call thin Nest services that delegate to domain query/command services.

8.2 audit

Attribute	Detail
Business purpose	Domain logic for **audit** (see `src/modules/audit/application/` use cases).
Folder	`src/modules/audit/` with typical layers: `application/`, `domain/`, `infrastructure/`, `presentation/`, `public/`.
HTTP (Nest)	`src/nest/audit/` Nest module + controller(s).
Factory	`createAuditModule` or `index.js` export in `src/modules/audit/`.
Persistence	PostgreSQL repositories under `src/modules/audit/infrastructure/` where present.
Tests	`tests/unit/**/audit*`, `tests/integration/**/audit*` (search by module name).

How it fits: Domain modules are composed in `src/app/bootstrap/create-application.js` and injected into Nest via `CoreDomainModule` (`src/nest/domain/`). Controllers call thin Nest services that delegate to domain query/command services.

8.3 authorization

Attribute	Detail
Business purpose	Domain logic for **authorization** (see `src/modules/authorization/application/` use cases).
Folder	`src/modules/authorization/` with typical layers: `application/`, `composition/`, `domain/`, `infrastructure/`, `presentation/`, `public/`.
HTTP (Nest)	`src/nest/authorization/` Nest module + controller(s).
Factory	`createAuthorizationModule` or `index.js` export in `src/modules/authorization/`.
Persistence	PostgreSQL repositories under `src/modules/authorization/infrastructure/` where present.
Tests	`tests/unit/**/authorization*`, `tests/integration/**/authorization*` (search by module name).

How it fits: Domain modules are composed in `src/app/bootstrap/create-application.js` and injected into Nest via `CoreDomainModule` (`src/nest/domain/`). Controllers call thin Nest services that delegate to domain query/command services.

8.4 cancellations

Attribute	Detail
Business purpose	Domain logic for **cancellations** (see `src/modules/cancellations/application/` use cases).
Folder	`src/modules/cancellations/` with typical layers: `application/`, `domain/`, `infrastructure/`, `presentation/`, `public/`.
HTTP (Nest)	`src/nest/cancellations/` Nest module + controller(s).
Factory	`createCancellationsModule` or `index.js` export in `src/modules/cancellations/`.
Persistence	PostgreSQL repositories under `src/modules/cancellations/infrastructure/` where present.
Tests	`tests/unit/**/cancellations*`, `tests/integration/**/cancellations*` (search by module name).

How it fits: Domain modules are composed in `src/app/bootstrap/create-application.js` and injected into Nest via `CoreDomainModule` (`src/nest/domain/`). Controllers call thin Nest services that delegate to domain query/command services.

8.5 channel-catalog-sync

Attribute	Detail
Business purpose	Domain logic for **channel-catalog-sync** (see `src/modules/integration-catalog-sync/application/` use cases).

Attribute	Detail
Folder	`src/modules/integration-catalog-sync/` with typical layers: `application/`, `domain/`, `infrastructure/`, `public/`.
HTTP (Nest)	No dedicated Nest folder; may be wired via compatibility or worker paths.
Factory	`createChannelCatalogSyncModule` or `index.js` export in `src/modules/integration-catalog-sync/`.
Persistence	PostgreSQL repositories under `src/modules/integration-catalog-sync/infrastructure/` where present.
Tests	`tests/unit/**/channel-catalog-sync*`, `tests/integration/**/channel-catalog-sync*` (search by module name).

How it fits: Domain modules are composed in `src/app/bootstrap/create-application.js` and injected into Nest via `CoreDomainModule` (`src/nest/domain/`). Controllers call thin Nest services that delegate to domain query/command services.

8.6 channels

Attribute	Detail
Business purpose	Domain logic for **channels** (see `src/modules/commerce-integrations/application/` use cases).
Folder	`src/modules/commerce-integrations/` with typical layers: `application/`, `domain/`, `infrastructure/`, `presentation/`, `public/`.
HTTP (Nest)	`src/nest/channels/` Nest module + controller(s).
Factory	`createChannelsModule` or `index.js` export in `src/modules/commerce-integrations/`.
Persistence	PostgreSQL repositories under `src/modules/commerce-integrations/infrastructure/` where present.
Tests	`tests/unit/**/channels*`, `tests/integration/**/channels*` (search by module name).

How it fits: Domain modules are composed in `src/app/bootstrap/create-application.js` and injected into Nest via `CoreDomainModule` (`src/nest/domain/`). Controllers call thin Nest services that delegate to domain query/command services.

8.7 compatibility

Attribute	Detail
Business purpose	Domain logic for **compatibility** (see `src/modules/compatibility/application/` use cases).
Folder	`src/modules/compatibility/` with typical layers: `application/`, `domain/`, `infrastructure/`, `presentation/`, `public/`.
HTTP (Nest)	No dedicated Nest folder; may be wired via compatibility or worker paths.

Attribute	Detail
Factory	`createCompatibilityModule` or `index.js` export in `src/modules/compatibility/`.
Persistence	PostgreSQL repositories under `src/modules/compatibility/infrastructure/` where present.
Tests	`tests/unit/**/compatibility*`, `tests/integration/**/compatibility*` (search by module name).

How it fits: Domain modules are composed in `src/app/bootstrap/create-application.js` and injected into Nest via `CoreDomainModule` (`src/nest/domain/`). Controllers call thin Nest services that delegate to domain query/command services.

8.8 external-id-mapping

Attribute	Detail
Business purpose	Domain logic for **external-id-mapping** (see `src/modules/external-id-mapping/application/` use cases).
Folder	`src/modules/external-id-mapping/` with typical layers: `application/`, `domain/`, `infrastructure/`, `public/`.
HTTP (Nest)	No dedicated Nest folder; may be wired via compatibility or worker paths.
Factory	`createExternalIdMappingModule` or `index.js` export in `src/modules/external-id-mapping/`.
Persistence	PostgreSQL repositories under `src/modules/external-id-mapping/infrastructure/` where present.
Tests	`tests/unit/**/external-id-mapping*`, `tests/integration/**/external-id-mapping*` (search by module name).

How it fits: Domain modules are composed in `src/app/bootstrap/create-application.js` and injected into Nest via `CoreDomainModule` (`src/nest/domain/`). Controllers call thin Nest services that delegate to domain query/command services.

8.9 identity

Attribute	Detail
Business purpose	Domain logic for **identity** (see `src/modules/identity/application/` use cases).
Folder	`src/modules/identity/` with typical layers: `application/`, `domain/`, `infrastructure/`, `presentation/`.
HTTP (Nest)	No dedicated Nest folder; may be wired via compatibility or worker paths.
Factory	`createIdentityModule` or `index.js` export in `src/modules/identity/`.
Persistence	PostgreSQL repositories under `src/modules/identity/infrastructure/` where present.
Tests	`tests/unit/**/identity*`, `tests/integration/**/identity*` (search by module name).

How it fits: Domain modules are composed in `src/app/bootstrap/create-application.js` and injected into Nest via `CoreDomainModule` (`src/nest/domain/`). Controllers call thin Nest services that delegate to domain query/command services.

8.10 inventory

Attribute	Detail
Business purpose	Domain logic for **inventory** (see `src/modules/inventory/application/` use cases).
Folder	`src/modules/inventory/` with typical layers: `application/` , `domain/` , `infrastructure/` , `presentation/` , `public/` .
HTTP (Nest)	`src/nest/inventory/` Nest module + controller(s).
Factory	`createInventoryModule` or `index.js` export in `src/modules/inventory/` .
Persistence	PostgreSQL repositories under `src/modules/inventory/infrastructure/` where present.
Tests	`tests/unit/**/inventory*` , `tests/integration/**/inventory*` (search by module name).

How it fits: Domain modules are composed in `src/app/bootstrap/create-application.js` and injected into Nest via `CoreDomainModule` (`src/nest/domain/`). Controllers call thin Nest services that delegate to domain query/command services.

8.11 marketplace-order-ingestion

Attribute	Detail
Business purpose	Domain logic for **marketplace-order-ingestion** (see `src/modules/marketplace-order-ingestion/application/` use cases).
Folder	`src/modules/marketplace-order-ingestion/` with typical layers: `application/` , `domain/` , `public/` .
HTTP (Nest)	No dedicated Nest folder; may be wired via compatibility or worker paths.
Factory	`createMarketplaceOrderIngestionModule` or `index.js` export in `src/modules/marketplace-order-ingestion/` .
Persistence	PostgreSQL repositories under `src/modules/marketplace-order-ingestion/infrastructure/` where present.
Tests	`tests/unit/**/marketplace-order-ingestion*` , `tests/integration/**/marketplace-order-ingestion*` (search by module name).

How it fits: Domain modules are composed in `src/app/bootstrap/create-application.js` and injected into Nest via `CoreDomainModule` (`src/nest/domain/`). Controllers call thin Nest services that delegate to domain query/command services.

8.12 marketplace-webhook-ingestion

Attribute	Detail
Business purpose	Domain logic for **marketplace-webhook-ingestion** (see `src/modules/marketplace-webhook-ingestion/application/` use cases).
Folder	`src/modules/marketplace-webhook-ingestion/` with typical layers: `application/`, `domain/`, `presentation/`, `public/`.
HTTP (Nest)	No dedicated Nest folder; may be wired via compatibility or worker paths.
Factory	`createMarketplaceWebhookIngestionModule` or `index.js` export in `src/modules/marketplace-webhook-ingestion/`.
Persistence	PostgreSQL repositories under `src/modules/marketplace-webhook-ingestion/infrastructure/` where present.
Tests	`tests/unit/**/marketplace-webhook-ingestion*`, `tests/integration/**/marketplace-webhook-ingestion*` (search by module name).

How it fits: Domain modules are composed in `src/app/bootstrap/create-application.js` and injected into Nest via `CoreDomainModule` (`src/nest/domain/`). Controllers call thin Nest services that delegate to domain query/command services.

8.13 marketplaces

Attribute	Detail
Business purpose	Domain logic for **marketplaces** (see `src/modules/marketplaces/application/` use cases).
Folder	`src/modules/marketplaces/` with typical layers: `application/`, `domain/`, `infrastructure/`, `presentation/`, `public/`.
HTTP (Nest)	`src/nest/marketplaces/` Nest module + controller(s).
Factory	`createMarketplacesModule` or `index.js` export in `src/modules/marketplaces/`.
Persistence	PostgreSQL repositories under `src/modules/marketplaces/infrastructure/` where present.
Tests	`tests/unit/**/marketplaces*`, `tests/integration/**/marketplaces*` (search by module name).

How it fits: Domain modules are composed in `src/app/bootstrap/create-application.js` and injected into Nest via `CoreDomainModule` (`src/nest/domain/`). Controllers call thin Nest services that delegate to domain query/command services.

8.14 mfa

Attribute	Detail
Business purpose	Domain logic for **mfa** (see `src/modules/mfa/application/` use cases).
Folder	`src/modules/mfa/` with typical layers: `application/`, `domain/`, `infrastructure/`, `presentation/`, `public/`.
HTTP (Nest)	`src/nest/mfa/` Nest module + controller(s).
Factory	`createMfaModule` or `index.js` export in `src/modules/mfa/`.
Persistence	PostgreSQL repositories under `src/modules/mfa/infrastructure/` where present.
Tests	`tests/unit/**/mfa*`, `tests/integration/**/mfa*` (search by module name).

How it fits: Domain modules are composed in `src/app/bootstrap/create-application.js` and injected into Nest via `CoreDomainModule` (`src/nest/domain/`). Controllers call thin Nest services that delegate to domain query/command services.

8.15 offers

Attribute	Detail
Business purpose	Domain logic for **offers** (see `src/modules/offers/application/` use cases).
Folder	`src/modules/offers/` with typical layers: `application/`, `domain/`, `infrastructure/`, `presentation/`, `public/`.
HTTP (Nest)	`src/nest/offers/` Nest module + controller(s).
Factory	`createOffersModule` or `index.js` export in `src/modules/offers/`.
Persistence	PostgreSQL repositories under `src/modules/offers/infrastructure/` where present.
Tests	`tests/unit/**/offers*`, `tests/integration/**/offers*` (search by module name).

How it fits: Domain modules are composed in `src/app/bootstrap/create-application.js` and injected into Nest via `CoreDomainModule` (`src/nest/domain/`). Controllers call thin Nest services that delegate to domain query/command services.

8.16 orders

Attribute	Detail
Business purpose	Domain logic for **orders** (see `src/modules/orders/application/` use cases).

Attribute	Detail
Folder	`src/modules/orders/` with typical layers: `application/`, `domain/`, `infrastructure/`, `presentation/`, `public/`.
HTTP (Nest)	`src/nest/orders/` Nest module + controller(s).
Factory	`createOrdersModule` or `index.js` export in `src/modules/orders/`.
Persistence	PostgreSQL repositories under `src/modules/orders/infrastructure/` where present.
Tests	`tests/unit/**/orders*`, `tests/integration/**/orders*` (search by module name).

How it fits: Domain modules are composed in `src/app/bootstrap/create-application.js` and injected into Nest via `CoreDomainModule` (`src/nest/domain/`). Controllers call thin Nest services that delegate to domain query/command services.

8.17 pricing

Attribute	Detail
Business purpose	Domain logic for **pricing** (see `src/modules/pricing/application/` use cases).
Folder	`src/modules/pricing/` with typical layers: `application/`, `domain/`, `infrastructure/`, `presentation/`, `public/`.
HTTP (Nest)	`src/nest/pricing/` Nest module + controller(s).
Factory	`createPricingModule` or `index.js` export in `src/modules/pricing/`.
Persistence	PostgreSQL repositories under `src/modules/pricing/infrastructure/` where present.
Tests	`tests/unit/**/pricing*`, `tests/integration/**/pricing*` (search by module name).

How it fits: Domain modules are composed in `src/app/bootstrap/create-application.js` and injected into Nest via `CoreDomainModule` (`src/nest/domain/`). Controllers call thin Nest services that delegate to domain query/command services.

8.18 products

Attribute	Detail
Business purpose	Domain logic for **products** (see `src/modules/products/application/` use cases).
Folder	`src/modules/products/` with typical layers: `application/`, `domain/`, `infrastructure/`, `presentation/`, `public/`.
HTTP (Nest)	`src/nest/products/` Nest module + controller(s).

Attribute	Detail
Factory	`createProductsModule` or `index.js` export in `src/modules/products/`.
Persistence	PostgreSQL repositories under `src/modules/products/infrastructure/` where present.
Tests	`tests/unit/**/products*`, `tests/integration/**/products*` (search by module name).

How it fits: Domain modules are composed in `src/app/bootstrap/create-application.js` and injected into Nest via `CoreDomainModule` (`src/nest/domain/`). Controllers call thin Nest services that delegate to domain query/command services.

8.19 returns

Attribute	Detail
Business purpose	Domain logic for **returns** (see `src/modules/returns/application/` use cases).
Folder	`src/modules/returns/` with typical layers: `application/`, `domain/`, `infrastructure/`, `presentation/`, `public/`.
HTTP (Nest)	`src/nest/returns/` Nest module + controller(s).
Factory	`createReturnsModule` or `index.js` export in `src/modules/returns/`.
Persistence	PostgreSQL repositories under `src/modules/returns/infrastructure/` where present.
Tests	`tests/unit/**/returns*`, `tests/integration/**/returns*` (search by module name).

How it fits: Domain modules are composed in `src/app/bootstrap/create-application.js` and injected into Nest via `CoreDomainModule` (`src/nest/domain/`). Controllers call thin Nest services that delegate to domain query/command services.

8.20 shipments

Attribute	Detail
Business purpose	Domain logic for **shipments** (see `src/modules/shipments/application/` use cases).
Folder	`src/modules/shipments/` with typical layers: `application/`, `domain/`, `infrastructure/`, `presentation/`, `public/`.
HTTP (Nest)	`src/nest/shipments/` Nest module + controller(s).
Factory	`createShipmentsModule` or `index.js` export in `src/modules/shipments/`.
Persistence	PostgreSQL repositories under `src/modules/shipments/infrastructure/` where present.
Tests	`tests/unit/**/shipments*`, `tests/integration/**/shipments*` (search by module name).

How it fits: Domain modules are composed in `src/app/bootstrap/create-application.js` and injected into Nest via `CoreDomainModule` (`src/nest/domain/`). Controllers call thin Nest services that delegate to domain query/command services.

8.21 tenants

Attribute	Detail
Business purpose	Domain logic for **tenants** (see `src/modules/tenants/application/` use cases).
Folder	`src/modules/tenants/` with typical layers: `application/` , `domain/` , `infrastructure/` , `presentation/` , `public/` .
HTTP (Nest)	`src/nest/tenants/` Nest module + controller(s).
Factory	`createTenantsModule` or `index.js` export in `src/modules/tenants/` .
Persistence	PostgreSQL repositories under `src/modules/tenants/infrastructure/` where present.
Tests	`tests/unit/**/tenants*` , `tests/integration/**/tenants*` (search by module name).

How it fits: Domain modules are composed in `src/app/bootstrap/create-application.js` and injected into Nest via `CoreDomainModule` (`src/nest/domain/`). Controllers call thin Nest services that delegate to domain query/command services.

8.22 webhooks

Attribute	Detail
Business purpose	Domain logic for **webhooks** (see `src/modules/webhooks/application/` use cases).
Folder	`src/modules/webhooks/` with typical layers: `application/` , `domain/` , `infrastructure/` , `presentation/` , `public/` .
HTTP (Nest)	`src/nest/webhooks/` Nest module + controller(s).
Factory	`createWebhooksModule` or `index.js` export in `src/modules/webhooks/` .
Persistence	PostgreSQL repositories under `src/modules/webhooks/infrastructure/` where present.
Tests	`tests/unit/**/webhooks*` , `tests/integration/**/webhooks*` (search by module name).

How it fits: Domain modules are composed in `src/app/bootstrap/create-application.js` and injected into Nest via `CoreDomainModule` (`src/nest/domain/`). Controllers call thin Nest services that delegate to domain query/command services.

Additional module-like areas (not separate top-level folders):

- `channel-catalog-sync` — catalog sync jobs and reconciliation scheduler (worker-heavy).

- `compatibility` — ChannelEngine / merchant compat read models and routes.
- `external-id-mapping` — integer ID mappings for compat APIs.
- `marketplace-order-ingestion` / `marketplace-webhook-ingestion` — inbound marketplace pipelines.

9 7. API endpoint reference

The authoritative list for **current Nest registrations** is in Wasl_Backend_Nest_Endpoint_Inventory.md (**159** routes).

Discovery limitations:

- Static regex scan may miss dynamically constructed paths or custom decorators.
- Global prefix (if any) is applied in Nest bootstrap — verify `configure-express.js` / `setGlobalPrefix` in `create-nest-application.js` when interpreting paths.
- Legacy `docs/API_ROUTE_INVENTORY.md` reflects Fastify-era scans (**not** authoritative post-migration).

Authentication patterns (verify per route):

- `@Public()` decorator or `isPublicRoute` — no auth.
- Default — `AuthGuard` requires JWT Bearer **or** API key headers.
- Merchant compat / StockConnect paths — additional rules in `auth.guard.js`.

10 8. Configuration and environment

All variables are declared in `src/app/config/schema.js` and loaded via `loadConfigFromEnvironment()`. Unknown keys are **ignored** (typo-safe deployments).

See Wasl_Backend_Environment_Variables.md for the complete name list.

Source	Role
`.env`	Local secrets (never commit)
`.env.example`	Documented template without production secrets
`docker-compose.yml`	Local Postgres/Redis/MinIO defaults
`--env-file=.env`	Used by `npm run dev/start` scripts

Sensitive variables (non-exhaustive): `DATABASE_URL`, `AUTH_JWT_*`, `AUTH_MFA_ENCRYPTION_KEY`, `STORAGE_SECRET_ACCESS_KEY`, API keys, marketplace connection secrets stored encrypted in DB.

StockConnect integration:

- `STOCK_CONNECT_INTEGRATION_ENABLED` — feature flag.
- `STOCK_CONNECT_BASE_URL`, `STOCK_CONNECT_SERVICE_TOKEN` — outbound/client configuration; **inbound** StockConnect API auth uses Wasl API keys (`nxk_` prefix), not the service token.

11 9. Database structure

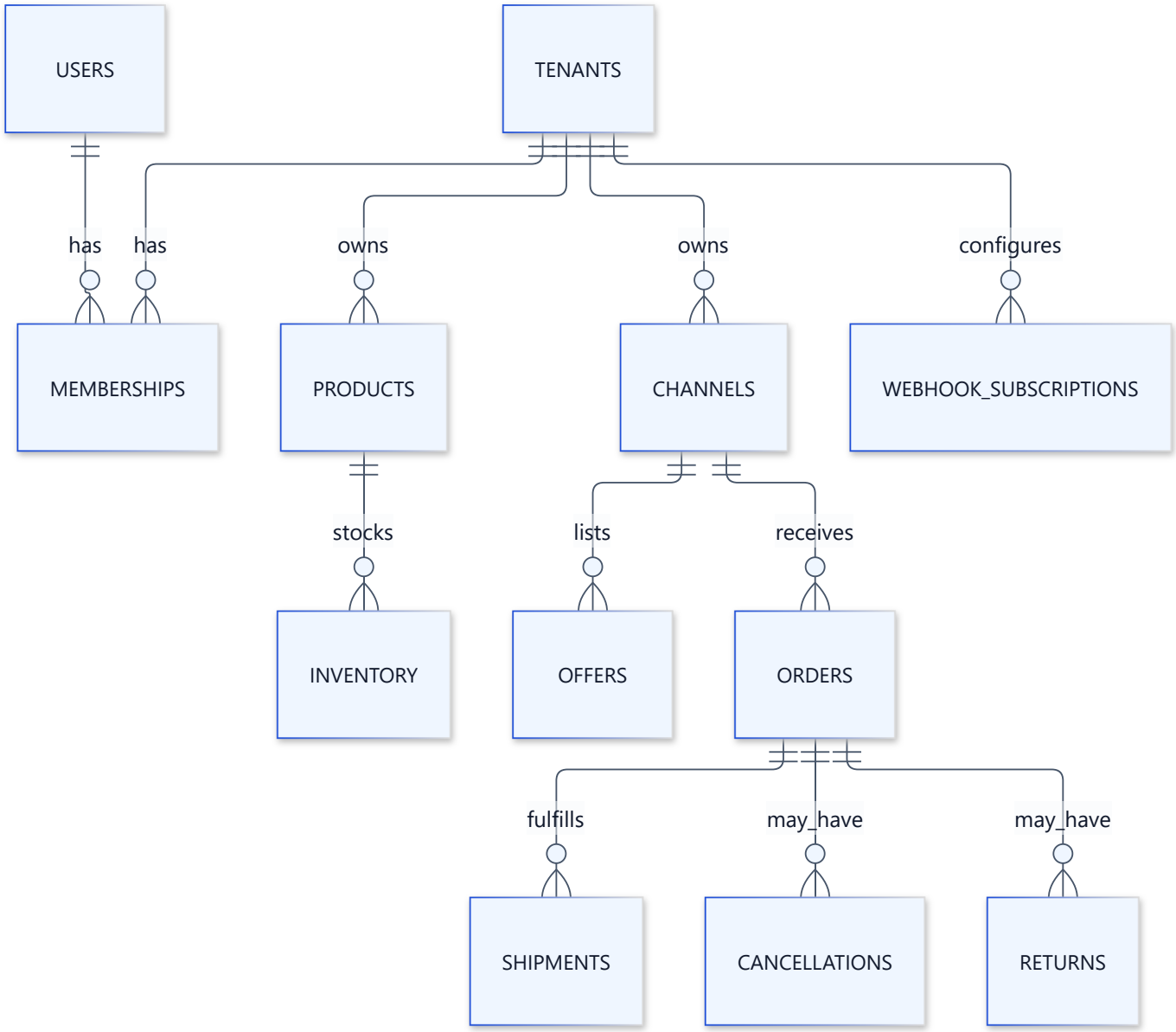
11.1 9.1 Engine and migrations

- **Engine:** PostgreSQL.
- **Migrations:** `47` files in `src/infrastructure/postgres/migrations/`, applied via `npm run migrate`.

11.2 9.2 Migration themes (chronological filenames)

Migration range	Domain
0001–0004	Schema, outbox, inbox, idempotency
0005–0010	Tenants, users, roles, auth, MFA
0011–0012	Audit, marketplaces
0013–0020	Channels, products, inventory, pricing, offers, commerce FKs
0023–0026	Orders, shipments, cancellations, returns
0027–0041	Permission backfills (phases 4–7)
0034–0035, 0044	Webhook subscriptions and deliveries
0042–0047	External ID mappings, marketplace connections, webhook ingress

11.3 9.3 ER diagram (logical — from migration names)



Caution: This diagram reflects **expected** relationships inferred from migration filenames and module boundaries, not an automated schema parse. Confirm columns/FKs in the SQL files before schema changes.

12 10. Testing and developer workflow

12.1 10.1 Commands (`package.json`)

Command	Purpose	Prerequisites
`npm install`	Install runtime + dev deps	Node 24+
`npm run dev`	API with watch + SWC	`.env` , Docker services
`npm run start`	API production-like	Same
`npm run dev:worker` / `start:worker`	Background worker	Redis, Postgres
`npm run build`	Syntax check all `src/**/*.js`	None
`npm run test:unit`	Vitest unit project	None
`npm run test:integration`	Vitest integration	Postgres, Redis, MinIO
`npm run test:jest`	Nest Jest suite	None
`npm run verify`	lint + arch + unit tests	None
`npm run migrate`	Apply DB migrations	`DATABASE_URL`
`docker compose up -d`	Local infra	Docker

Note: Integration tests were **not** re-executed as part of generating this document unless separately noted in your CI logs.

12.2 10.2 Tracing a feature

1. Find route in Wasl_Backend_Nest_Endpoint_Inventory.md.
2. Open matching `*.controller.js` → `*.service.js` .
3. Follow call into `src/modules/<domain>/application/*` .
4. Inspect `infrastructure/postgres-*-repository.js` for SQL tables touched.
5. Search `tests/integration/*` for the route or use case name.

13 11. Deployment and infrastructure

13.1 11.1 Docker

- `Dockerfile` — API container image (inspect file for Node version and start command).
- `docker-compose.yml` — Postgres **5433**, Redis **6380**, MinIO **9000/9001** on host; observability profile optional.

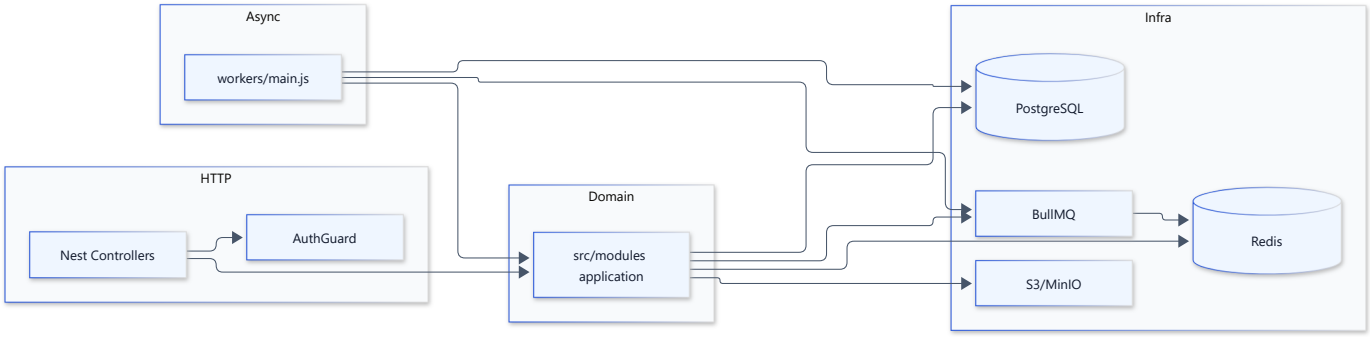
13.2 11.2 CI/CD

No `.github/workflows` were found in this repository snapshot — pipeline may live elsewhere or be not yet committed.

13.3 11.3 Production configuration

Use `NODE_ENV=production`, strong `AUTH_JWT_*` or asymmetric keys, real `DATABASE_URL`, dedicated Redis with `noeviction`, and locked-down `SERVER_CORS_ORIGINS`.

14 12. Dependency and architecture map



15 13. Architecture risks and documentation gaps

Finding	Evidence	Impact
Dual HTTP route representations	Nest controllers vs legacy `*.routes.js`; stale `audit-route-inventory.mjs`	Wrong route docs if using old inventory
ESLint skips `src/nest/**`	`eslint.config.js`	Nest decorator files not linted by default
Build syntax check skips Nest	`scripts/check-syntax.mjs`	Nest files rely on SWC/runtime compile
Large surface area, uneven test depth	220+ integration test files but not all modules may have equal coverage	Regressions possible in untested paths
RLS and tenant context	Postgres RLS migrations	Bugs in context setup cause data leaks or denials — requires careful review
BullMQ Redis policy	`docker-compose.yml` documents `noeviction`	Wrong Redis config drops jobs silently
Unverified runtime execution	This guide generation	Execute `test:integration` before release

Separate **confirmed defects** from **risks**: items above are risks/documentation gaps unless linked to a failing test or ticket.

16 14. Glossary

Term	Meaning
Tenant	Organization slice; data isolated via tenant_id + RLS
Channel	Sales channel linked to marketplace connection
Offer	Product listing on a channel with pricing
Outbox	Transactional event table for reliable async dispatch
Inbox / idempotency	Dedup for webhooks and retries
Core domain	Object graph injected as `CORE_DOMAIN` into Nest
Merchant compat	ChannelEngine-style query-parameter API compatibility
StockConnect	Integration API under `/api/v2/stock-connect`

17 15. Coverage appendix

17.1 15.1 File inventory

- **CSV rows (excl. header):** 1171
- **Paths under `src/`:** 925
- **Layer breakdown:** aggregate via CSV pivot on `layer` column.

17.2 15.2 Endpoints documented

- **Nest static scan:** 159

17.3 15.3 Dependencies reviewed

- **Count:** 42

17.4 15.4 Diagrams in this document

1. Bootstrap sequence (§4.1)
2. HTTP auth flow (§5.1)
3. BullMQ sequence (§5.3)
4. Startup outbox/listen (§5.7)
5. Logical ER (§9.3)
6. Architecture map (§12)

17.5 15.5 Regeneration

```
node docs/architecture/build-architecture-guide-companions.mjs
node docs/architecture/generate-architecture-guide-md.mjs
node docs/architecture/generate-architecture-guide-pdf.mjs
```

End of Wasl Backend Complete Architecture and Technology Guide.